UAV Emergency Rescue System for Disaster Response

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1 · Introduction

Earthquakes and floods destroy the terrestrial networks survivors rely on, exactly where rescue demand peaks. In the 72-hour golden window, time is spent locating victims rather than reaching them. BeiDou-3 Short Message Communication (BDS-

2 · Objectives

- *RTK-GNSS positioning to centimetre accuracy as the system position source*
- *Autonomous UAV mission control via QGroundControl / PX4 (MAVLink)*
- *YOLO-based survivor detection with precise object positioning*

3 · Methods

A simulation-first stack (ROS 2 + Gazebo + PX4 SITL + QGroundControl) integrates five work packages. An earthquake Gazebo world models collapsed buildings and blocked roads with all terrestrial networks [illegible] leaving BDS-3 short messaging as the only channel. Modules are built independently, then merged [illegible] communication backbone encodes an RTK-corrected coordinate into a fixed-point 112-bit binary payload (lat, lon at 7 dp, altitude, uncertainty radius, triage priority, survivor ID), transmits it through the BeiDou regional service, decodes it at a ground station, and publishes it on a latched /target/emergency_coordinate topic that mission-planning and ground-control modules consume. Delivery reliability and latency are measured in a field campaign and reported with Wilson confidence bounds.

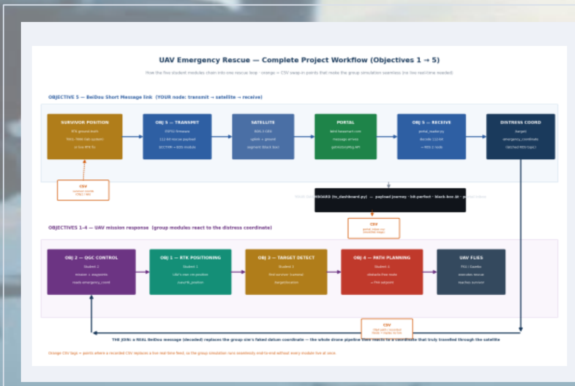

Figure 1. End-to-end rescue chain across the five objectives (O1 RTK → O3 detect → O4 plan → O5 BeiDou → O2 mission).

4 · Results

The decoded coordinate drives an autonomous ROS 2 mission trigger, merged and verified across the five-module system. The BeiDou link delivered all 232 field transmissions across open sky, light canopy, urban canyon and indoor (Wilson 95%

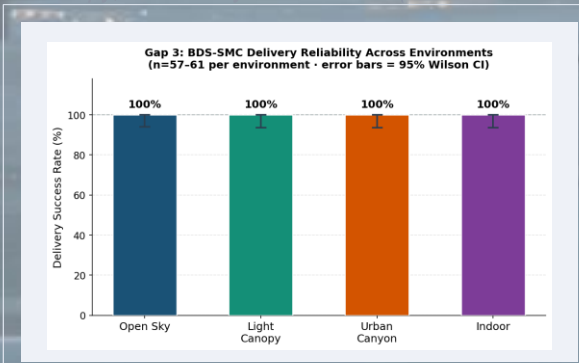

Figure 2. BeiDou delivery reliability by environment with Wilson 95% intervals (n = 57–61 per environment).

5 · Discussion

Among infrastructure-free links, only satellite systems survive a disaster: COSPAS-SARSAT carries no user-definable triage payload and

6 · Conclusions

A complete survivor record fits one 112-bit BeiDou message and decodes losslessly. The link delivered 232/232 transmissions (Wilson ≥ 93.7%).

7 · Future Work

- *Physical satellite transmission of the 112-bit frame*
- *Diurnal latency characterisation on the operational payload*

References

[1] China Satellite Navigation Office. BeiDou-3 RDSS / Short Message Service ICD.
[2] COSPAS-SARSAT. T.001 — 406 MHz Distress Beacon Specification.
[3] BeiDou short message [illegible] Space: Science & Technology, 2022.
[4] PX4 Autopilot [illegible] documentation, 2024.

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